

SWAYAM NAKTE

Computer Engineering Student

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🌐 [Github](#) [LinkedIn](#) [Portfolio](#)

Education

K J Somaiya School of Engineering

Bachelor Of Technology, Computer Engineering

2023 – 2027

CGPA: 8.9/10

Skills

Languages: C++, Python, JavaScript, Go

Frameworks/Libraries: React, Node.js, Mongoose, Microservices

Databases & Cloud: MongoDB, PostgreSQL, MySQL, Redis, Docker, AWS, Git/GitHub

Web & Mobile: HTML, CSS, REST APIs, WebSockets, Flutter, Dart

Auth, State & Data Tools: Redux Toolkit, OAuth 2.0, JWT

DevOps & Tools: GitHub Actions, Postman

Projects

Aquila | High-Performance Quantitative Backtesting Framework | *C++23, CRTP, std::pmr, RDTSC* **Jul 2026**

- Built a lock-free SPSC event bus with cache-line isolation (1.15 ns/op, 2.2x faster than std::queue) for higher event throughput with less latency jitter, and a std::pmr arena allocator cutting allocation from 41 ns to 14 ns
- Replaced virtual dispatch with CRTP and Structure-of-Arrays layout using 2MB huge pages, cutting TLB entries for 1M-bar datasets from 2048 to 4 and enabling SIMD-vectorized SMA at 4.76 ns/op
- Implemented a full Signal-to-Order-to-Fill backtest loop with slippage, commission, Welford's $O(1)$ rolling standard deviation (Bollinger Bands), and TSC profiling measuring p50 = 40 ns dispatch latency

RunSphere | GPS Territory-Capture Running Platform | *TypeScript, React Native, Node.js, PostgreSQL, PostGIS* **ongoing**

- Engineered closed-loop GPS detection (thinned $O(n^2)$ scan, spherical-excess area) and a spatial carving pipeline with Uber H3 hexagonal indexing for $O(n)$ contested-territory resolution and BFS-pruned residual cells
- Implemented a two-stage ML anti-cheat pipeline: a 19-feature kinematic extractor feeding an Isolation Forest scorer with a 3-tier decision policy and cold-start gating
- Developed the full-stack mobile app in React Native (Expo) with live GPS recording, offline-first idempotent sync, and MapLibre territory rendering, backed by a Node.js/PostgreSQL API

CodeMortem | Real-Time Competitive Programming Arena | *Go, Next.js, Microservices, Redis, Docker, Azure* **Feb 2026**

- Built a real-time competitive programming platform with live 1v1 matches, a learn track of 50+ problems, and 25 lessons; sustained 99.9% uptime over 4 weeks since launch
- Engineered a high-concurrency Go WebSocket Hub sustaining 250+ concurrent 1v1 matches at sub-ms broadcast latency, with Glicko-2 matchmaking and dynamic rating-window expansion
- Built a hybrid evaluation engine (local bypass + Judge0) with concurrent Goroutines, driving verdict latency under 50ms; shipped 50+ CI/CD deployments to a 4-container Azure stack via GitHub Actions

Experience

Software Development Cell, KJSSE

Jan 2026 – June 2026

Full Stack Developer Intern

K.J. Somaiya School of Engineering, Mumbai

- Developed a full-stack MERN alumni portal managing 10,000+ alumni records dating back to 2007, replacing fragmented manual tracking systems
- Built and maintained 30+ RESTful API endpoints covering alumni profiles, event management, and resource sharing, with optimized MongoDB queries achieving sub-150ms average response times
- Collaborated in a 2-person cross-functional team using Agile methodology with weekly sprints, independently leading full-stack development while integrating datasets processed by a teammate; manually tested and validated all endpoints using Postman

Additional Information

Competitive Programming

- [Codeforces](#) Specialist (1594), [CodeChef](#) 3-Star (1750), [Leetcode](#) Knight (1875)
- 2100+ problems solved across different coding platforms
- Secured Rank 1050 among 25,000+ participants in [Educational Codeforces Round 192](#)

Awards/Activities: Abacus State Level Runner-up

Languages: English, Hindi, Marathi

Certifications: [Google Cloud Cybersecurity](#), [CodeChef Github](#), [CodeChef Machine Learning](#)